Ex. No: 14 File Allocation Strategies

PROGRAM A: SEQUENTIAL FILE ALLOCATION

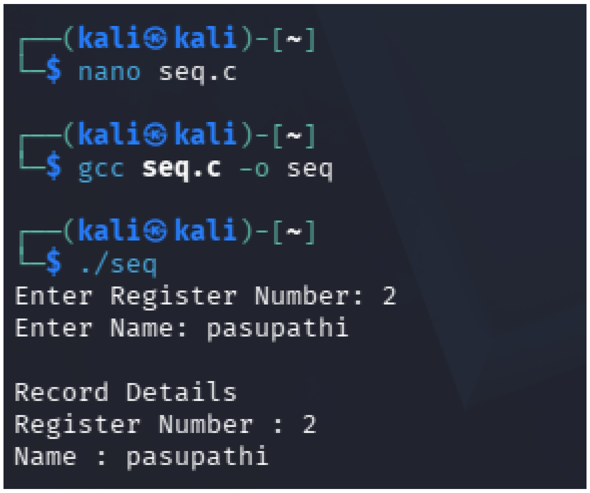
C PROGRAM

```
#include <stdio.h>
int main()
{
int start, length, i;
printf("Enter Starting Block: ");
scanf("%d", &start);
printf("Enter File Length (Number of Blocks): ");
scanf("%d", &length);
printf("\nAllocated Blocks:\n");
for(i = 0; i < length; i++)
{
printf("%d ", start + i);
```



```
}
printf("\n");
return 0;
}
```

Output:



```
┌──(kali㉿kali)-[~]
└─$ nano seq.c

┌──(kali㉿kali)-[~]
└─$ gcc seq.c -o seq

┌──(kali㉿kali)-[~]
└─$ ./seq
Enter Register Number: 2
Enter Name: pasupathi

Record Details
Register Number : 2
Name : pasupathi
```

SHELL SCRIPT

```
#!/bin/bash

echo "Enter Starting Block:"

read start

echo "Enter File Length:"

read length

echo "Allocated Blocks:"

for ((i=0;i<length;i++))

do

echo -n "$((start+i)) "

done

echo
```

Output:

```
┌──(kali㉿kali)-[~]
└─$ nano lab14-1.sh

┌──(kali㉿kali)-[~]
└─$ chmod +x lab14-1.sh

┌──(kali㉿kali)-[~]
└─$ bash lab14-1.sh
Enter Starting Block:
10
Enter File Length:
5
Allocated Blocks:
10 11 12 13 14
```

PROGRAM B: INDEXED FILE ALLOCATION

C PROGRAM

```
#include <stdio.h>

int main()

{

int n, indexBlock, blocks[20], i;

printf("Enter Index Block: ");

scanf("%d", &indexBlock);

printf("Enter Number of Blocks: ");

scanf("%d", &n);

printf("Enter Block Numbers:\n");
```

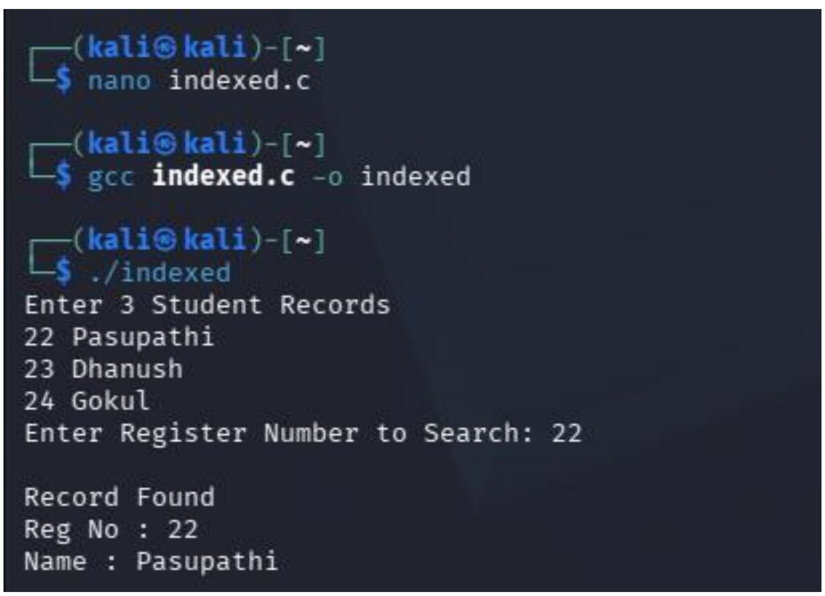


```
for(i = 0; i < n; i++)

{
```

```
scanf("%d", &blocks[i]);
}
printf("\nIndex Block : %d\n", indexBlock);
printf("Allocated Blocks : ");
for(i = 0; i < n; i++)
{
printf("%d ", blocks[i]);
}
printf("\n");
return 0;
}
```

Output:

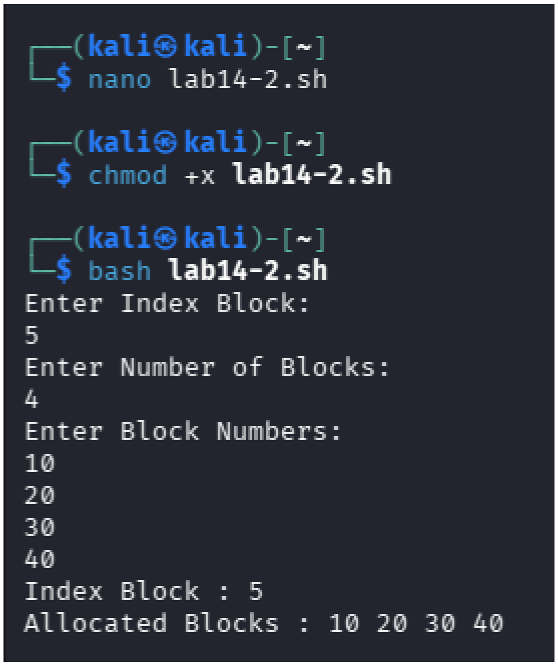


SHELL SCRIPT

```
#!/bin/bash
echo "Enter Index Block:"
```

```
read index
echo "Enter Number of Blocks:"
read n
echo "Enter Block Numbers:"
for ((i=0;i<n;i++))
do
read block[$i]
done
echo "Index Block : $index"
echo -n "Allocated Blocks : "
for ((i=0;i<n;i++))
do
echo -n "${block[$i]} "
done
echo
```

Output:



PROGRAM C: LINKED FILE ALLOCATION

C PROGRAM

```
#include <stdio.h>

int main()

{

int n, blocks[20], i;

printf("Enter Number of Blocks: ");

scanf("%d", &n);

printf("Enter Block Numbers:\n");

for(i = 0; i < n; i++)

{

scanf("%d", &blocks[i]);

}

printf("\nLinked Allocation:\n");

for(i = 0; i < n - 1; i++)

{
```

printf("%d --> ", blocks[i]);

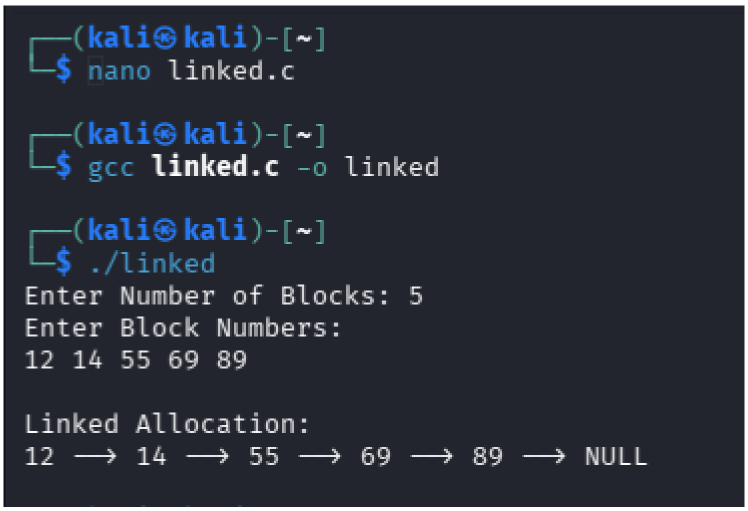
}

printf("%d --> NULL\n", blocks[n - 1]);

return 0;

}

Output:



SHELL SCRIPT

#!/bin/bash

echo "Enter Number of Blocks:"

read n

echo "Enter Block Numbers:"

for ((i=0;i<n;i++))

do

read block[$i]

done

echo "Linked Allocation:"

for ((i=0;i<n-1;i++))

do

```
echo -n "${block[$i]} --> "

done

echo "${block[$((n-1))]} --> NULL"
```

Output: